Casey-Named Principle Candidate — Commitment vs Boundary: the geometry commits, the boundary dresses

Casey Koons (mechanism) · written up by Keeper · 2026-07-11 · Candidate for Cal ratification.

Statement

The geometry predicts *commitments*, not *dressings*. A particle's *commitment* is its discrete, conserved content — the quantum numbers (charge, spin, color), the loading, and the bare/Lagrangian (current) mass. The geometry fixes these exactly. A particle's *dressing* is the continuous boundary energy added on emission into 3D — the QCD glue, the confinement cloud. Mass, as measured in a *dressed* form (the constituent mass), is a secondary, sheddable, emission-threshold quantity:

> The commitment is the conserved skeleton; the mass is the sheddable balloon. The bare mass the geometry commits is the *current* mass; the constituent mass is current + the boundary dressing.

Consequences (why it earns naming)

1. It says which mass the geometry predicts: the current (Lagrangian) mass, not the constituent. The geometry commits the bare value; the ~300 MeV QCD glue dresses it to constituent. *This is the target ruling that carries F506* — the down-quark ratios land against current masses (s/d=20 at 0.5%), and Friday's "miss" was comparing the bare commitment to the *dressed* (constituent) value.
2. It explains the two-tier precision structure. Commitments (discrete, conserved) → exact predictions (charge, spin, θ_QCD, topology). Boundary dressings (continuous, blurred, scheme-dependent) → the ~10^{-2} floor (masses). Not a limitation — the difference between the conserved skeleton and the sheddable balloon.
3. It explains why the mass sector is universal in the mean, blurred in the width, conditional at the extreme. The commitment is geometry-set (universal); the dressing has an intrinsic width (the blur) and shifts in extreme media (quark-gluon plasma, neutron-star cores).
4. It grounds the neutrino as the bookkeeping remainder. In decay, the commitment (conserved quantum numbers) rides across to the daughter; the excess boundary energy is shed — a chunk carried by the neutrino (the boundary's remainder). Heavy-quark decay = downshift + emit, energy on the neutrino.
5. It sets the emission-threshold reading of mass. Mass is what emission costs; emission has a threshold (Wallach k_min = N_c), not a fixed value — so mass is conditional (extreme media shift it), universal in the cold-dilute regime because the threshold is geometry-set.

Discipline / bar for ratification

- The "compare to current, not constituent" rule must be applied consistently and in advance, not chosen per-sector to make things land. F506 is the test case: current is forced by *this principle*, not by matching.
- The principle does NOT excuse a miss as "just the dressing" — the dressing must be a *named, computed* quantity (the QCD glue, ~300 MeV), not an open fudge (the standing discipline from the discrete/continuous principle).
- It is a *framework organizing principle*, not an observable bank; it explains the shape of the mass sector and fixes the target, not the numbers.

Relation to existing principles

Refines the discrete/continuous principle (Casey's) at the particle level: information = commitment (discrete, exact); energy = dressing (continuous, blurred). Pairs with the membrane-refraction mechanism (Wallach = the Snell's law; the mass is the refracted/committed component) and the two-tier precision hypothesis (Elie).

— Keeper, 2026-07-11. Candidate: *the geometry commits (discrete, conserved — charge, spin, the current mass), exactly; mass in dressed form is a secondary emission-threshold quantity — the sheddable boundary energy — uni-*

versal in the mean, blurred in the width, conditional at the extreme. Load-bearing now: it fixes the current-mass target that carries F506. For Cal ratification.